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Game-day temperatures are predictive of National Football League game outcomes when teams from different climates compete against each other

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ABSTRACT

NFL teams may have an advantage over their opposition when players are more familiar with game-day temperatures; however, relationships between game-day air temperature and team performance had not been examined using official at-stadia air temperatures. Game data from 2017 to 25 were collected from NFL "Gamebooks." Data were extracted for games contested by a team based in a cool climate >39° N (North teams, $n = 17$) against a team based in a warm climate <39° N (South teams, $n = 15$). Data extracted included game-day temperature (°C), game location (North, South), game outcome (win, no-win [relative to North team]), and point differential (points [relative to North team]). Mixed effects logistic regressions examined whether game-day temperature associated with game outcome, while mixed effects models examined whether temperature associated with point differential. Models were run for all games combined, and for games contested in North and South locations, respectively. North team win probability declined as game-day temperature increased (OR = 0.974, 95% CI [0.951, 0.997], $p = .026$), and North team point differential was negatively associated with temperature (β = -0.175, 95% CI [−0.286, −0.063], $p = .002$). Models run according to game location revealed that North team point differential was associated with game-day temperature for games played South (β = -0.152, 95% CI [−0.242, −0.063], $p = .001$). In NFL games contested by North (>39° N) versus South (<39° N) teams, North teams are less likely to win and have worse point differentials, as game-day temperatures increase. Research should explore whether behavioral or psychological adaptations to familiar game-day temperatures explain these findings.

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American football; climate; home advantage; performance; sport

Introduction

The National Football League (NFL) is an American football competition contested by 32 teams across the United States. All teams are based in northern latitudes, with the southernmost Miami Dolphins located at 25.76° N, and the northernmost Seattle Seahawks located at 47.61° N. Average annual air temperatures are lower at higher latitudes, such that from 2017 to 24, the combined mean (± SD) annual temperature for NFL cities located above 39° N was 11 (±2)°C, compared with 19 (±3)°C for NFL cities located below 39° N [1]. These differential temperatures are unlikely to induce physiological adaptations that confer an advantage at given game-day temperatures; for example, while heat acclimatization improves *aerobic* performance in the heat [2], it does not usually affect performance of high-intensity 5–6-second exercise bouts, as is characteristic of American football [3]. However, this does not preclude players from developing *behavioral* (e.g. cooling strategies in the heat, warming strategies in the cold) or *psychological* (e.g. reduced perceptual response to cold or hot temperatures) adaptations that may optimize performance in familiar environmental conditions [4,5]. While such adaptations at the individual-level may play a role in creating a "home advantage" for teams, they may also benefit teams when competing away from home in familiar game-day temperatures.

Some studies support the hypothesis that NFL teams have an advantage over their opposition when players are more familiar with game-day temperatures [6,7]. From 2002 to 20, teams based in cold climates above 39° N had a home field win percentage of 62% during the winter months (Dec-Feb); a higher home field win percentage than the league average of 57% recorded during winter,

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and across the season [6]. An earlier 2007 study also found that NFL teams had an advantage over their opposition when game-day temperatures more closely reflected those experienced across the preceding 5 days [7]. Despite these findings, previous studies have not used actual game-day temperatures recorded at NFL stadia to model relationships between game-day temperature and game outcomes. Nor have studies examined game point differentials (win/loss margin), alongside overall game outcome (win probability), to determine the *magnitude* of any benefit conferred to teams at given game-day temperatures. And finally, there is a need to explore these questions using more recent historical NFL game data to ensure findings are relevant and can inform current-day practice.

For this study, data from eight NFL seasons (2017–25) were collated to examine whether game-day air temperature is associated with NFL team win probability and point differential when teams based in cold climates (>39° N) compete against teams from warm climates (<39° N). It was hypothesized that – irrespective of game location – win probability and point differential would associate with game-day air temperature, such that when competing against teams based <39° N, teams from northern latitudes >39° N would be less likely to win and have worse point differentials, as game-day temperatures increase.

Materials and methods

Historical NFL game data across eight seasons (2017–25) were collected from "Gamebooks" published on the NFL website. These documents provide official statistics for NFL games. Gamebooks were collected for all regular-season and playoff games contested by a team based >39° N, and a team based <39° N. Of the 32 NFL teams, $n = 17$ have a home stadium located >39° N and were assigned to the "North" group, while $n = 15$ have a home stadium located <39° N and were assigned to the "South" group. The 39° N was used to categorize teams into North or South group as this study aimed to build on findings of previous research that used this latitude [5]. Data obtained from the National Weather Service website revealed that average annual temperature from 2017 to 24 for NFL team cities in the North group was 11°C (range 7–14°C), while for NFL team cities in the South group annual temperature averaged 19°C (range 14–28°C). Data extracted from Gamebooks for each eligible game included at-stadia game-day air temperature (°C) and humidity (%), as well as game location (North, South). For games played at indoor stadiums, reported temperature and humidity data inside the stadium were used. Further, game data, relative to the North team, were recorded, including game outcome (win, no-win), and point differential (points). Each team was also assigned an identification number (e.g. 49ers = 01 and Bears = 02) and all seasons were coded (e.g. 2017–18 = 1 and 2018–19 = 2). For the game outcome variable, which was used to examine the odds of *winning*, tied games were coded as "no win" because both competing teams (incl. North team) fail to win when a game ends tied. The present study was deemed exempt from ethical review, and approved, by the Deakin University Human Research Ethics Committee due to use of preexisting non-identifiable data (ID: 2023–011).

To test the hypothesis that North teams are disadvantaged as game-day temperatures increase, mixed effects logistic regression models with robust standard errors were used to examine whether maximum game-day temperature was associated with game outcome, while mixed effects models were used to examine whether game-day temperature was associated with point differential. Models were run for all games combined, and separately for games contested in North and South locations, respectively. All initial models included "game-day temperature" as a single fixed effect, with follow-up models including game-day "humidity" and "location" (models with all games combined only) as fixed effects. Random effects in all models included "season year" (2017–18 = season 1 etc.), "home team ID," and "rank differential" (difference in end-season table position between competing teams). These were included to somewhat control for any inherent differences in team quality between competing teams. When modeling game outcome, results show the odds ratio of North teams winning for each 1°C increase in game-day temperature. When modeling point differential, results show the mean change in game point differential, relative to the North team, for each 1°C increase in game-day temperature. Models used the maximum likelihood estimation method with an unstructured covariance matrix and robust standard errors. Linear regression models with clustered standard errors (by NFL city) were also used to compare game-day temperatures between North and South games. Statistical significance was set at $p < 0.05$, and

95% confidence intervals (CI) reported. Analyses were conducted using Stata Statistical Software (Release 17, College Station, TX: StataCorp LLC).

Results

Across eight NFL seasons (2017–2025), a total of 1047 gamebooks were collated for North vs. South team games. Game-day temperature was not reported for 36 games, leaving 1011 games for analyses. Of these, 526 (52%) were contested in North (> 39° N) locations and 485 (48%) in South (< 39° N) locations. From North teams' perspective, 97.5% games contested in North locations were played at the home stadium of the competing North team (2.5% international: London [10 games], Munich [2 games], Frankfurt [1 game]), while 99.2% games contested in South locations were played away at the home stadium of the competing South team (0.4% international: Mexico City [2 games]; 0.4% Super Bowl [2 games]). Game-day temperature averaged 16.5°C (95% CI: 15.9, 17.1°C) overall, with average temperature lower ($p < 0.001$) for games played in North locations (12.0°C, 95% CI: 11.2, 12.8°C) compared with games played in South locations (21.3°C, 95% CI: 20.7, 22.0°C). Of the 1011 analyzed games, North teams won 545 games (54%), including 309 games (59%) in North locations and 236 games (49%) in South locations.

North team win probability declined as game-day temperature increased (OR = 0.974, 95% CI [0.951, 0.997], $p = .026$, Figure 1). Models remained statistically significant when game-day humidity (OR = 0.974, 95% CI [0.951, 0.998], $p = .033$) and game location (OR = 0.980, 95% CI [0.962, 0.999], $p = .036$) were included as covariates. Data show odds of North team winning declines ~2–3% for each one-degree (C) increase in game-day temperature. When data were modeled according to game location, change in North team win probability as game-day temperature increased did not reach statistical significance for games played North (OR = 0.981, 95% CI [0.951, 1.011], $p = .211$, Figure 1) or South (OR = 0.979, 95% CI [0.958, 1.002], $p = .071$, Figure 1). These findings were unchanged when game-day humidity was included as a covariate in models (North games, OR = 0.979, 95% CI [0.948, 1.010], $p = .185$; South games, OR = 0.980, 95% CI [0.960, 1.002], $p = .072$).

North team point differential was negatively associated with game-day temperature (β = −0.175, 95% CI [−0.286, −0.063], $p = .002$, Figure 2). Models remained statistically significant when game-day humidity was included as a covariate (β = −0.182, 95% CI [−0.294, −0.071], $p = .001$) but not when game location was included (β = −0.109, 95% CI [−0.230, 0.012], $p = .078$). These data indicate that each 1°C increase in game-day temperature associates with worsened point differentials (i.e. winning margin decreases or losing margin increases) of between 0.1 and 0.2 points. When data were modeled according to game location, change in North team point differential as game-day temperature increased did not reach statistical significance for games played North (β = −0.082, 95% CI [−0.243, 0.079], $p = .318$, Figure 2) but did for games played South (β = −0.152, 95% CI [−0.242, −0.063], $p = .001$, Figure 2). When game-day humidity was included as a covariate in models, temperature was not statistically significant for games played North (β = −0.104, 95% CI [−0.271, 0.064], $p = .225$) but remained significant for games played South (β = −0.149, 95% CI [−0.236, −0.062], $p = .001$).

Discussion

This study of 1011 games across eight NFL seasons (2017–25) revealed that when North teams based >39° N competed against South teams based <39° N, North team win probability declined, and point differential worsened, as game-day air temperature increased. This remained true for win probability even when accounting for game location (North, South) as a potential confounder. Models run according to game location revealed relationships between point differential and game-day temperature only reached statistical significance for games played South, which suggests overall findings were driven mostly by outcomes of games played in warmer conditions.

Across all 1011 games, models indicate that North team win probability decreased 2–3% for each 1°C increase in game-day temperature. Previous studies of elite soccer and Australian football performances have similarly shown higher game-day temperatures improve odds of winning for teams based in warm climates when competing against teams from cooler climates [8,9]. While the present findings suggest that familiarity with game-day temperature likely *contributes* to the phenomenon of "home advantage"

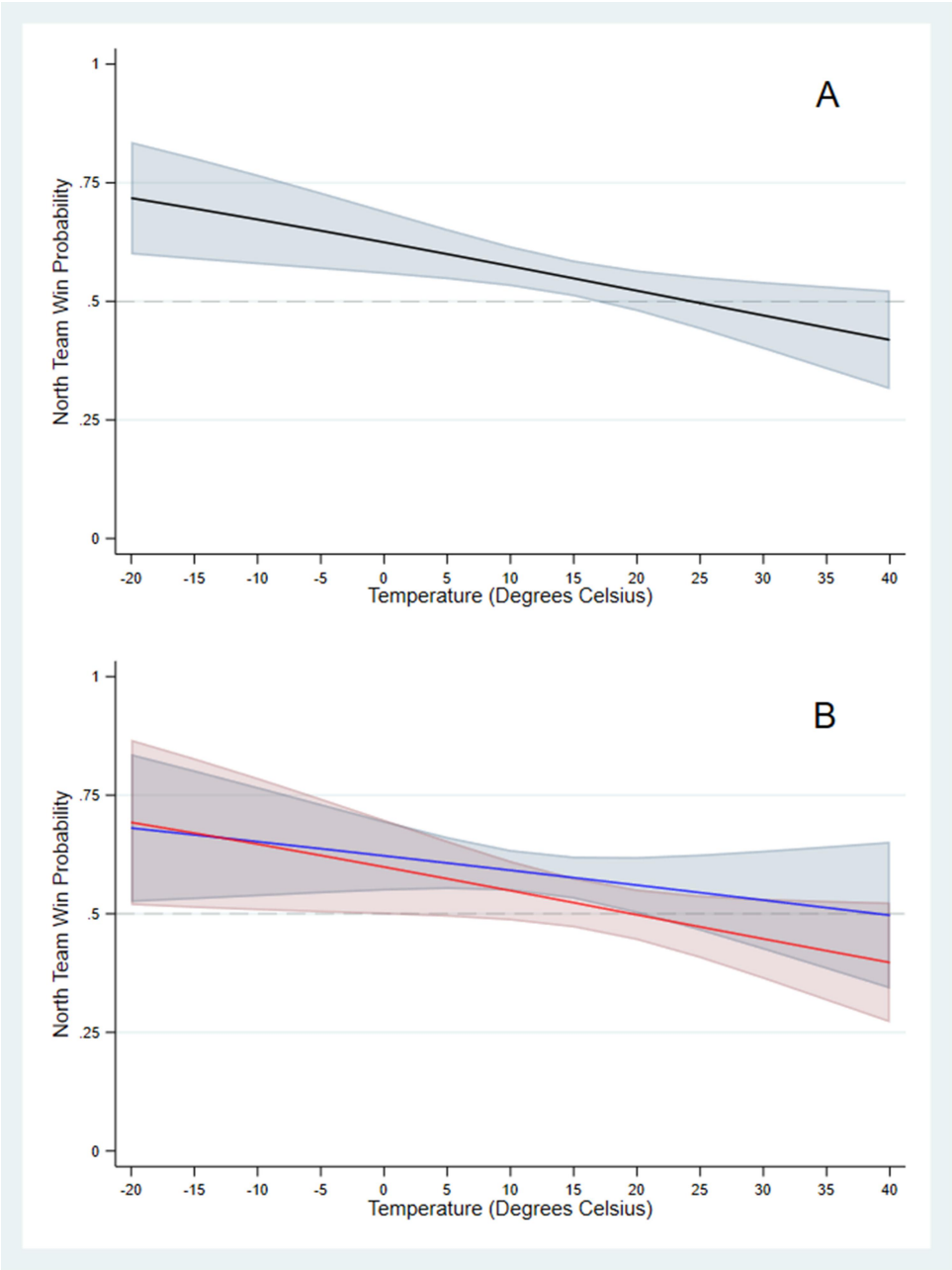

Figure 1. North team win probability as game-day temperature increases for all games ($n = 1{,}011$) contested against South teams (panel A), and for games played in North ($n = 526$, blue data) and South ($n = 485$, red data) locations (panel B). Data derived from eight NFL seasons (2017–2025). Modelled win probability for given temperatures (lines) shown with 95% confidence intervals (shaded).

[10], the association between game-day temperature and win-probability was still statistically significant (OR = 0.974 vs OR = 0.980) when including "game location" as a model covariate. This supports the notion that game-day temperature may affect team performance *independent* of other travel-related factors, such as partisan crowds and stadium or playing surface familiarity [10]. In this context, it is perhaps surprising that the association between game-day temperature and *point differential* did not remain statistically significant when "game location" was included as a covariate. One possible explanation for this may be that the capacity of players to adapt and perform at high or low game-day temperatures may be especially important for games with narrow point differentials, in which the result is uncertain throughout the game, and which may produce greater demands (physical, cognitive) on players. For games with larger point differentials and where the result is certain, the final scores of each

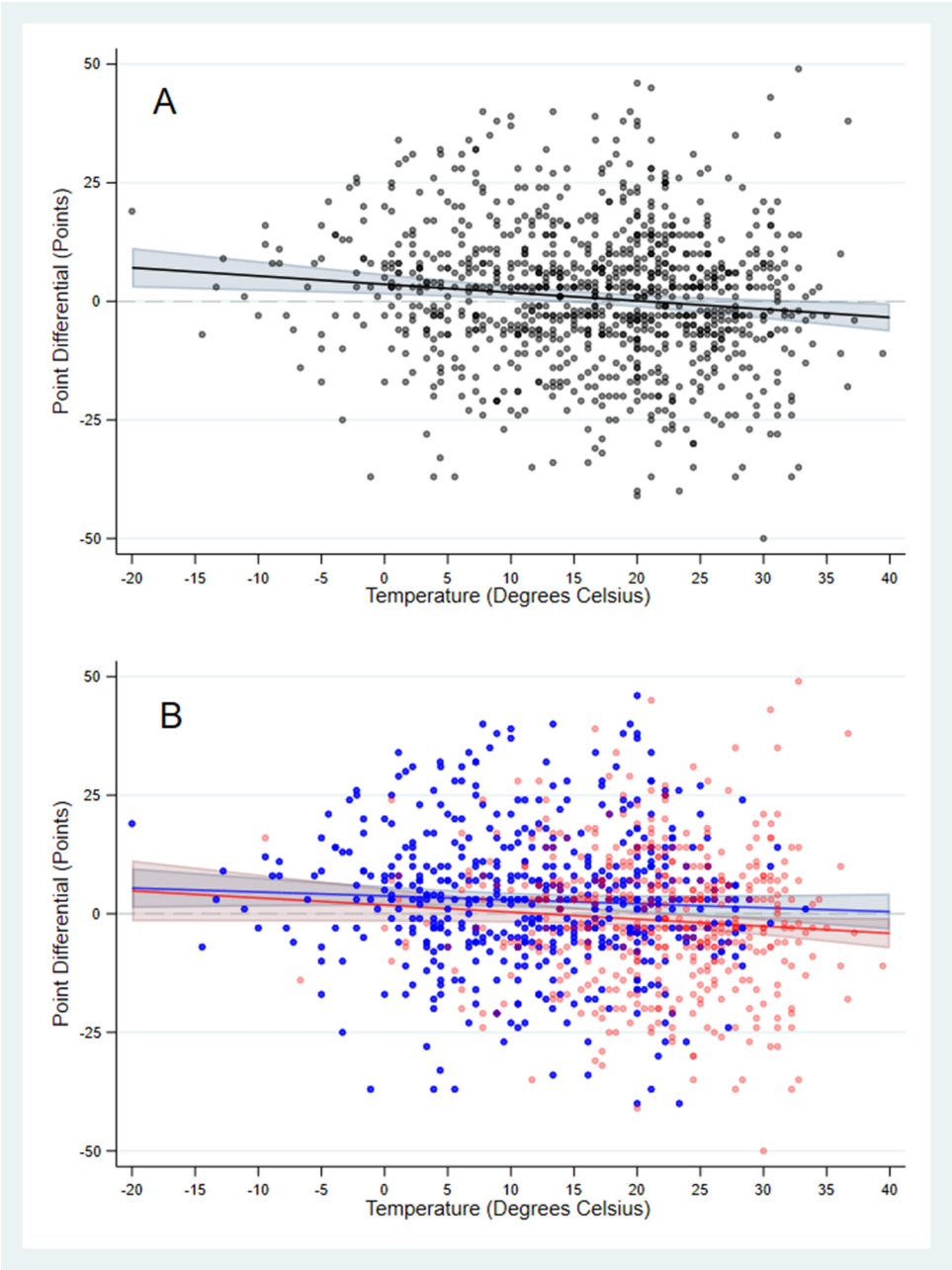

Figure 2. North team point differential as game-day temperature increases for all games ($n = 1{,}011$) contested against South teams (panel A), and for games played in North ($n = 526$, blue data) and South ($n = 485$, red data) locations (panel B). Data derived from eight NFL seasons (2017–2025). Modelled point differential for given temperature (lines) shown with 95% confidence intervals (shaded).

team may be less influenced by players' response to game-day temperatures due to lower overall game demands/player effort. Support for the notion that physical demand/effort is lower when score differentials are high can be found in a study of professional soccer players, which found that when games had larger goal differentials, players covered less distance and had lower acceleration/deceleration loads [11].

The underlying mechanism(s) behind the present findings cannot be elucidated from this study due the observational study design and absence of data on players' physiological, behavioral, and psychological responses to different game-day temperatures. While there is good evidence that heat acclimatization can improve subsequent aerobic performance in the heat, evidence for a benefit to explosive power-based exercise performance, as typically required in American football, is lacking [2,12]. As such, we speculate that

the present findings may reflect behavioral and/or psychological adaptations players develop with exposure to warm or cold temperatures. Indeed, given the change in North team point differential as game-day temperature increased reached significance for games played South ($\beta = -0.152$, $p = .001$), and that there was also a trend toward significance when win probability was modeled for games played South (OR = 0.979, $p = .071$), it is possible that players and/or team staff familiar with warmer conditions (i.e. South teams) may develop behaviors or strategies that optimize performance in such conditions. For example, *hypo*hydration can impair sport-specific skill execution [13], and pre-game intra-venous *hyper*hydration is a relatively common practice among NFL teams and players prior to games played in the heat [14]. If this practice was more common among players from South teams based <39° N, it may confer an advantage when competing in the heat and explain, to some extent, the current findings. In-game cooling strategies (e.g. ice ingestion, ice/cold towels applied to skin, and increased air flow to wet skin) can also improve thermal comfort and lower the rectal temperatures of athletes during sporting competition [15,16], and such strategies are potentially optimized by players or staff more familiar with hot conditions. These explanations, however, are speculative and should be tested by future research.

Players may also adapt to competing in cold conditions, and such adaptations may also partially explain the present findings [6]. A "cold weather advantage" for NFL teams has previously been proposed in a study that revealed teams based in cold climates >39° N had a higher win percentage during the winter months (62%) than the league average both during winter (57%) and across the whole season (57%) [6]. Potential adaptations to cold that may improve sport performance are not well understood; however, when exercise economy was assessed in college American football players undertaking outdoor practice (mean temperature during winter practice = 0°C) and compared to that of athletes doing similar amounts of indoor training (e.g. basketball), so-called cold acclimatized American footballers had better exercise economy (lower VO_2 uptake) at submaximal workloads than indoor athletes when exercising in cold (5°C) conditions [17]. Other research in non-athletes suggests that repeated cold stress may improve cold tolerance, reducing shivering and vasoconstrictive responses, leading to energy conservation and improved thermal comfort, which may benefit sport performance [5]. It has even been suggested that such adaptations may alleviate cold-induced cognitive impairments, although supportive data for this are limited [5].

Evidence here that game-day temperatures associate with win-probability when North teams based >39° N compete against South teams based <39° N suggests a benefit to team performance when NFL players are familiar with game-day temperatures. Research should explore the efficacy of behavioral and/or psychological strategies (e.g. hydration and cooling strategies) aimed at improving players' readiness and performance at anticipated game-day temperatures. The present findings also have potential implications for the integrity of the NFL competition. For example, the Superbowl is hosted by a different city each year, but cities located South (< 39° N) have historically hosted more Superbowls (22/26 since the year 2000). This may reflect a preference for favorable game-day weather; however, findings here suggest that Superbowls hosted in the South, particularly in warm conditions, favor South teams when the Superbowl is a North vs South team contest.

This study has limitations. First, the use of the 39° N latitude to categorize teams is somewhat arbitrary for identifying teams familiar with cold or warm temperatures; however, this line was selected to build on findings of previous research that also used 39° N to explore relationships between climate and NFL team performance [5]. Furthermore, of teams located close (±1°) to 39° N (i.e. Washington, Baltimore, Cincinnati, Kansas City, Indianapolis, and Denver), all except one (Washington) were North teams; therefore, most games in our analysis were contested by teams with a significant differential in latitude and climate, as evidenced by the 11°C mean difference in game-day temperature between North and South locations. Findings here now encourage future, more nuanced, examinations of how NFL team location and home climate influence performance. Second, data collated was limited to games completed across eight seasons from years 2017 to 2025. While this restricted the sample of games analyzed, this was done to ensure data could be extracted from official "gamebooks" published by the NFL, and to ensure findings were driven by *recent* game outcomes and so had current-day relevance. Third, when modeling relationships between temperature and game outcomes, it is difficult to control for the inherent quality of competing teams for any given game; therefore, findings may have been influenced by consistent discrepancies in the quality of North and South teams at given temperatures. To somewhat address this, we included "season year," "team ID" and "end-season rank differential" as random effects in models. Given this approach, and the large sample of 1011

games, it is unlikely that consistent differences in team quality significantly influenced findings. Fourth, we did not control for home time-zone/circadian misalignment between competing teams in our analyses, although such misalignment could affect game outcomes. This decision was made to minimize random effects and overfitting of models. Finally, the methods used to record game-day temperature at NFL games have not been made public; therefore, the extent to which these were consistent across stadiums, or reflect temperatures experienced by players throughout a given game, is not clear. Nonetheless, data used in the present study were reported as part of "game-day weather" conditions in official NFL gamebooks. Moreover, when games were played at indoor stadiums, both indoor and outdoor weather conditions were reported in NFL Gamebooks, with indoor conditions used in analyses.

This study of historical data across eight NFL seasons (2017–25) shows that when North teams based >39° N compete against South teams based <39° N, North teams are less likely to win and have worse point differentials (win by less, or lose by more), as game-day air temperatures increase. Importantly, the finding that win probability associates with game-day temperature remains true even when accounting for game-location ("home advantage") in statistical models. Findings of this study are mostly driven by results of games played in the south, where game-day air temperatures are typically warmer than in the north. Future research should explore possible explanations for these findings, including via lab- and field-based studies examining the efficacy of behavioral and psychological strategies aimed at improving player and team performances across different game-day temperatures.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

Data used for this study are publicly available via Gamebooks published on the National Football League website (https://www.nfl.com/scores/) but can also be provided upon request by contacting the corresponding author.

Code availability statement

Code used for this study is available as supplementary material.

Ethics approval statement

This study was deemed exempt from ethical review, and approved, by the Deakin University Human Research Ethics Committee due to use of preexisting publicly available data (ID: 2023–011).

Author contributions

S.R. and S.W. conceptualized the study and methodology. S.R., J.R., and C.U. extracted the data. S.B. and S.R. conducted statistical analyses. S.R. and C.U. prepared manuscript drafts. S.W., J.R., and S.B. reviewed and edited manuscript drafts.

Abbreviations

NFL	National Football League
° N	Degrees of Latitude North of Equator
°C	Degrees Celsius (temperature)
OR	Odds Ratio
VO_2	Volume of oxygen consumption

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